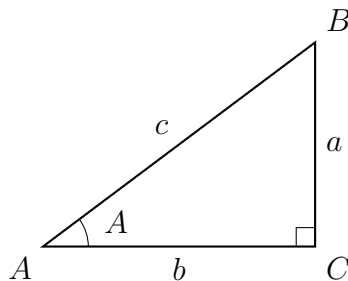


Pythagorean Theorem: Which Side Goes Where?

Finding a hypotenuse, finding a missing leg, and finding coordinate distance without dropping the hypotenuse into a leg slot

Directions:

- $a^2 + b^2 = c^2$ holds only when c is the **hypotenuse** — the side across from the right angle, and always the longest side.
- Before you compute, write down which given side is the hypotenuse. If the hypotenuse is known and a leg is missing, you *subtract*: $b^2 = c^2 - a^2$.
- Sanity-check every answer: a leg must come out *shorter* than the hypotenuse.
- On a **find the mistake** problem, name the mistake in words *and* show correct work. Round decimals to the nearest hundredth.



How every triangle here is labelled. No values shown: use the numbers given in each problem.

1. A right triangle has legs of 6 cm and 8 cm. Find the length of its hypotenuse.

Answer: _____ cm

2. A right triangle has a hypotenuse of 13 cm and one leg of 5 cm. Find the length of the other leg.

Answer: _____ cm

3. Find the mistake. A 17-foot ladder leans against a wall with its foot 8 feet from the wall. Jae wants the height h the ladder reaches and writes:

$$17^2 + 8^2 = h^2 \quad h \approx 18.79 \text{ feet}$$

(a) Explain how you can tell Jae's answer is impossible *before* checking any arithmetic. (b) Name the substitution mistake, then find the correct height.

Answer: _____ ft

4. Find the distance between $A(2, 3)$ and $B(7, 15)$ on the coordinate plane. Sketch the right triangle whose hypotenuse is $\overline{AB}$.

Answer: _____ units

5. **Find the mistake.** A right triangle has legs of 9 cm and 40 cm. Sam writes: “*40 is the longest side I was given, so it must be the hypotenuse,*” and computes

$$\sqrt{40^2 - 9^2} \approx 38.97 \text{ cm}$$

(a) Explain why 40 cannot be the hypotenuse here. (b) Find the correct hypotenuse.

Answer: _____ cm

6. Find the mistake. Kim finds the distance from $P(-3, 2)$ to $Q(5, 8)$. She writes: “The horizontal difference is 8 and the vertical difference is 6. The 8 is bigger, so it is the hypotenuse,” and computes

$$\sqrt{8^2 - 6^2} \approx 5.29 \text{ units}$$

(a) Name Kim’s mistake — which segment really is the hypotenuse? (b) Find the correct distance PQ .

Answer: _____ units

7. A television screen is a rectangle whose diagonal measures 65 inches and whose height is 33 inches. Its width x satisfies

$$x^2 + 33^2 = 65^2$$

(a) Solve the equation for *every* value of x . (b) Which solution is the screen’s width, and why must the other one be thrown out?

Answer: _____

8. A rectangular field measures 80 m by 60 m. Rosa walks from one corner to the opposite corner along two sides of the field. Tam walks the straight diagonal between the same two corners.

How many meters shorter is Tam's walk than Rosa's?

Answer: _____ m